

University of Yaoundé I
Department of Physics, Faculty of Science
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

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Dear Editors,
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We are pleased to submit our manuscript, “**Scaffold-Controlled Evaluation of Molecular Representations for Antimalarial Activity Prediction**”, for consideration as a Research Article in the *Journal of Computer-Aided Molecular Design*.

The manuscript presents a scaffold-controlled comparison of graph, sequence, and topological molecular representations against an ECFP4-random-forest reference. We test this question on a frozen panel of 19 836 African antimalarial natural products under random and Bemis–Murcko scaffold cross-validation (25 fold–seed replicates), comparing GIN, GIN–TFP, GIN–TNE, and ChemBERTa with ECFP4–RF.

Two contributions matter for readers of JCAMD. First, the evaluation makes **fold-independent transformer initialization** explicit: pretrained ChemBERTa weights are restored before fine-tuning in every fold, removing the cross-fold weight-transfer channel that can inflate apparent transformer performance under cross-validation. This protocol detail is rarely reported in molecular benchmarks and is required for a valid transformer comparison. Second, the topological-fusion arms (GIN–TFP, GIN–TNE) test persistent-homology and tensor-network descriptors inside a graph architecture under scaffold-held-out evaluation on an antimalarial natural-product panel. The molecular benchmark is the primary analysis; a phenotype-only LISH reference is provided as orthogonal context in the Supporting Information (Section S1).

Under the scaffold split, **ECFP4–RF achieves the highest ROC AUC (0.830 0)**, ahead of GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7) and ChemBERTa (0.786 7); all four learned arms are significantly lower after multiplicity correction. Under the random split the ordering is unchanged (ECFP4–RF 0.943 3; all paired $p < 0.000 1$ after Benjamini–Hochberg correction). Representation scale alone does not guarantee predictive gains on this panel. The topological fusion arm narrows the gap modestly, while projection-weight analysis identifies persistent-image dimensions as a salient block; this is presented as descriptive evidence about representation use, not causal attribution.

As an external molecule-disjoint transfer analysis, we repeated the ECFP4–RF versus GIN comparison on a molecule-disjoint ChEMBL-derived *P. falciparum* activity panel ($n = 22\,267$ after exclusion of 180 canonical-SMILES overlaps). ECFP4–RF again outperformed GIN under the random (0.954 7 versus 0.923 7) and scaffold (0.919 0 versus 0.884 3) splits. This extends the comparison beyond the internal screening labels, while the ChEMBL labels remain thresholded IC_{50}/EC_{50} records rather than prospective experiments. The benchmark supports a bounded conclusion: under the evaluated panels, split protocols, and model configurations, the tested learned representations did not improve on ECFP4–RF. We present this as a dataset- and protocol-bounded result, not as a statement about learned molecular models in general. Code, frozen splits and trained checkpoints are available in the public repository (https://github.com/NanaEngo/Malaria_codesV2).

This work is original, is not under consideration elsewhere, all authors have approved the submission, and the authors declare no competing financial interests.

We thank you for your consideration.

Myke Vital Sao Temgoua
Corresponding Author